

L39 — B-Trees: Structure and Operations

Unit: 3 | Chapter: Trees | BT Level: BT6 | CO: CO2, CO4

Learning Objectives

- Define B-tree and its structural properties
 - Insert and delete keys in a B-tree with splitting and merging
 - Explain why B-trees are preferred for disk-based storage
 - Distinguish B-trees from B+ trees and binary trees
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1. Motivation: Disk-Based Data Structures

For in-memory data, AVL/Red-Black trees with $O(\log_2 n)$ performance are excellent.

For **disk-based storage** (databases, file systems):

- Disk reads are **1000** $\times$ **slower** than memory reads
- Each "disk access" reads a fixed-size **page** (e.g., 4KB)
- Goal: minimize the number of disk accesses = minimize tree height

B-tree: A generalization of BST where each node can contain **many keys** and have **many children**, reducing tree height drastically.

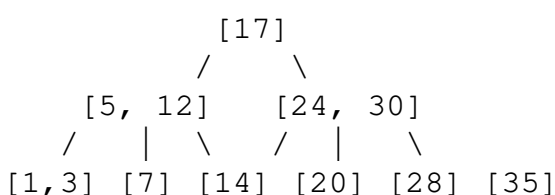
2. B-Tree Definition (Order t)

A B-tree of minimum degree t ($t \geq 2$) satisfies:

1. Every node has at most **$2t - 1$ keys** (max $2t$ children)
2. Every non-root node has at least **$t - 1$ keys** (min t children)
3. All leaves have the same depth (perfectly balanced)
4. Keys in each node are **sorted**
5. For any key $k[i]$ in a node, all keys in $\text{child}[i] < k[i] < \text{all keys in child}[i + 1]$

B-tree of order $t = 2$ (2-3-4 tree): Nodes have 1–3 keys, 2–4 children.

3. B-Tree Structure ($t = 2$)



- Height $h \approx \log_t(n) \rightarrow$ very shallow
 - Each node is a disk page $\rightarrow$ few disk accesses per operation
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4. Node Representation

```
class BTreeNode:
    def __init__(self, leaf=False):
        self.keys = []          # List of keys (sorted)
        self.children = []      # List of child nodes
        self.leaf = leaf        # True if this is a leaf node

class BTree:
    def __init__(self, t):
        self.t = t              # Minimum degree
        self.root = BTreeNode(leaf=True)
```

5. Search

```
def search(self, k, node=None):
    """Search for key k. Returns (node, index) or None."""
    if node is None:
        node = self.root

    i = 0
    while i < len(node.keys) and k > node.keys[i]:
        i += 1

    if i < len(node.keys) and k == node.keys[i]:
        return (node, i)    # Found!

    if node.leaf:
        return None         # Not found

    return self.search(k, node.children[i])    # Recurse into child
```

Trace: search(14) in tree above:

1. Root [17]: $14 < 17 \rightarrow$ go to child[0] = [5, 12]
 2. [5, 12]: $14 > 12 \rightarrow$ go to child[2] = [14]
 3. [14]: $14 == 14 \rightarrow$ **Found!**
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6. Insertion

Key idea: Insert always at a leaf. If a node becomes **overfull** ($2t$ keys), **split** it by pushing the median key up to the parent.

6.1 Split a Full Child

```
def _split_child(self, parent, i):
    """Split the i-th child of parent (which is full with  $2t-1$  keys)."""
    t = self.t
```

```

full_child = parent.children[i]
new_child = BTreeNode(leaf=full_child.leaf)

# Median key
mid = t - 1

# Move median to parent
parent.keys.insert(i, full_child.keys[mid])
parent.children.insert(i + 1, new_child)

# Split keys and children
new_child.keys = full_child.keys[mid + 1:]
full_child.keys = full_child.keys[:mid]

if not full_child.leaf:
    new_child.children = full_child.children[t:]
    full_child.children = full_child.children[:t]

```

6.2 Insert Key

```

def insert(self, k):
    root = self.root

    if len(root.keys) == 2 * self.t - 1:    # Root is full → split
        new_root = BTreeNode(leaf=False)
        new_root.children.append(self.root)
        self._split_child(new_root, 0)
        self.root = new_root

    self._insert_non_full(self.root, k)

def _insert_non_full(self, node, k):
    i = len(node.keys) - 1

    if node.leaf:
        # Insert key in sorted position
        node.keys.append(None)
        while i >= 0 and k < node.keys[i]:
            node.keys[i + 1] = node.keys[i]
            i -= 1
        node.keys[i + 1] = k
    else:
        # Find appropriate child
        while i >= 0 and k < node.keys[i]:
            i -= 1
        i += 1

        if len(node.children[i].keys) == 2 * self.t - 1:
            self._split_child(node, i)
            if k > node.keys[i]:
                i += 1

        self._insert_non_full(node.children[i], k)

```

7. Insertion Trace ($t = 2$)

Initial B-tree: Root = [10, 20, 30, 40] (full, $2t-1 = 3$ means max 3 keys; here $t = 2$)

Inserting 25:

1. Root [10, 20, 30] is not full ($t = 2$, max = 3 keys) $\rightarrow$ go to appropriate child
2. 25 is between 20 and 30 $\rightarrow$ go to middle child
3. Insert 25 in leaf $\rightarrow$ leaf now has [25] $\rightarrow$ Done

Inserting 35 (causes split): If leaf [30, 35, 40] becomes full (3 keys = $2t-1$):

- Median = 35
- Push 35 up to parent
- Split: left child [30], right child [40]

8. Deletion (Overview)

Deletion is more complex — key may be in internal node or leaf.

Three cases:

1. **Key in leaf:** Delete directly (if node has $\geq t$ keys after deletion)
2. **Key in internal node:** Replace with in-order predecessor/successor; then delete from leaf
3. **Node underflows ($< t-1$ keys):**
 - **Borrow** from a sibling (rotation through parent)
 - **Merge** with sibling + parent key (reduces parent's keys by 1)

9. B-Tree vs B+ Tree

Feature	B-Tree	B+ Tree
Data stored	In all nodes	Only in leaf nodes
Leaf linking	No	Leaves linked (for range queries)
Space efficiency	Lower	Higher (internal = index only)
Range queries	Slower	Fast (scan linked leaves)
Use in databases	Less common	Standard (MySQL InnoDB, PostgreSQL)

10. Complexity Analysis

Operation	Time
Search	$O(t \cdot \log_t n)$
Insert	$O(t \cdot \log_t n)$
Delete	$O(t \cdot \log_t n)$

Height $O(\log_t n)$

For $t = 500$ (typical disk page size), a B-tree with 1 billion keys has height $\approx \log_{500}(10^9) \approx 3$ levels $\rightarrow$ only **3 disk reads** to find any key!

Summary

- B-tree of degree t : each node has $t-1$ to $2t-1$ keys; perfectly balanced; all leaves at same depth.
 - **Search, insert, delete:** $O(t \cdot \log_t n)$ — very shallow, minimizes disk I/O.
 - **Insert:** Split full nodes proactively on the way down (or on overflow).
 - **Delete:** Merge or borrow from siblings to maintain minimum key count.
 - B+ trees (standard in databases) store data only in leaves, linked for range scans.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 18 (MIT Press, 2022)
- Ramakrishnan & Gehrke, *Database Management Systems*, Ch. 9 (McGraw-Hill, 2002)